Math for Economists Bootcamp

Module 2: Calculus and optimization

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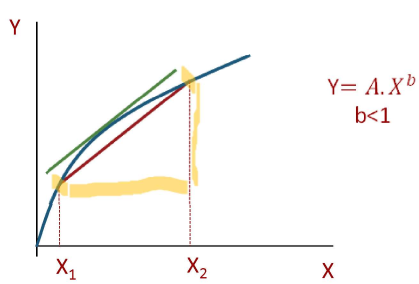
Derivative

- **Derivative** – the slope of the tangent line to a function at a point, calculated by taking the limit of the difference quotient, is the derivative



- $y = f(x)$
- $\frac{\Delta y}{\Delta x} = \frac{f(X_2) - f(X_1)}{(X_2 - X_1)}$
- $\frac{\Delta y}{\Delta x} = \frac{f(X_1 + \Delta x) - f(X_1)}{\Delta x}$
- $\lim_{\Delta x \to 0} \frac{f(X_1 + \Delta x) - f(X_1)}{\Delta x}$
- Example: $y = mx + b$
  - $\lim_{\Delta x \to 0} \frac{m(X_1 + \Delta x) + b - mX_1 - b}{\Delta x}$
  - $\lim_{\Delta x \to 0} \frac{(m\Delta x)}{\Delta x} = m$

- **Differentiation** – the process of taking a derivative

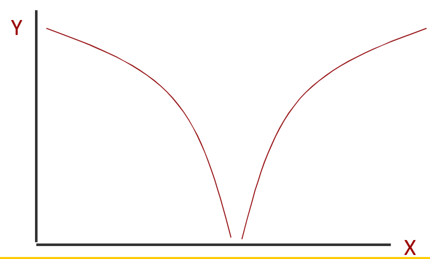
Calculus rules

- *If c is a constant:* $\frac{dc}{dx}$*=0*
- *If c is a constant: f: c*f(x)* $\rightarrow f'(x) = c * f'(x)$
- $f(x) = mx^n + c$ $\rightarrow f'^{(x)} = m.n.x^{n-1}$
- $y(u)$, where $u = f(x)$ $\rightarrow \frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$ (chain rule)
- $y = e^x \rightarrow \frac{dy}{dx} = e^x$
- $y = \ln(x) \rightarrow \frac{dy}{dx} = \frac{1}{x}$
- $h(x) = g(x) + f(x)$ $\rightarrow h'(x) = g'(x) + f'(x)$
- $h(x) = \sum_i g_i(x)$ $\rightarrow h'(x) = \sum_i g_i'(x)$
- $h(x) = g(x).f(x)$ $\rightarrow h'(x) = g'(x).f(x) + f'(x).g(x)$
- $h(x) = g(x)/f(x)$ $\rightarrow h'(x) = \frac{g'(x).f(x) - f'(x).g(x)}{[f(x)]^2}$

Chain Rule - the chain rule defines the derivative of a composite function as the derivative of the outer function evaluated at the inner function times the derivative of the inner function

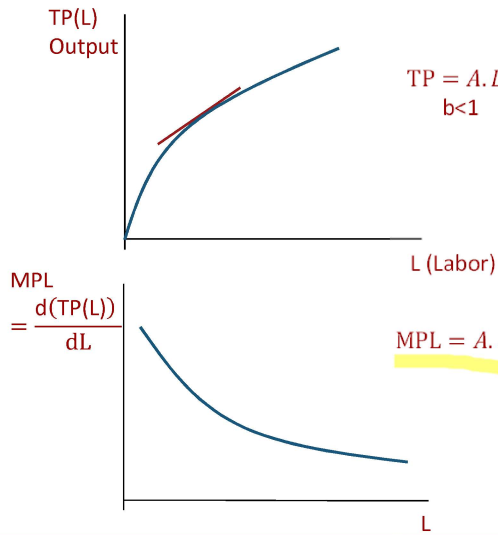
Existence, Uniqueness and solutions

- In economics we usually use differentiable functions
- **Differentiable function** - a function for which f ′(x) exists
- If $f(x)$ is a continuous function then x is a non-empty and closed set; and if bounded then there will be both a minimum and a maximum



Derivatives and Economics

- Single variable
  - $f(x) = ax^b$
    - $f'(x) = \frac{df(x)}{dx} = a.b.x^{b-1}$



$\mathrm{TP} = A.L^b$

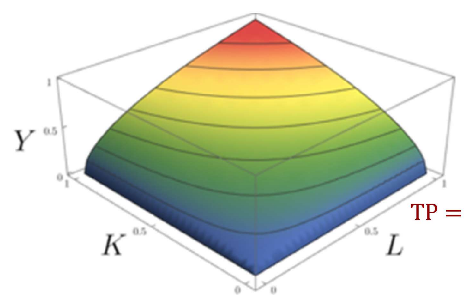
b<1

$\mathrm{MPL} = A.b.L^{b-1}$

Derivative

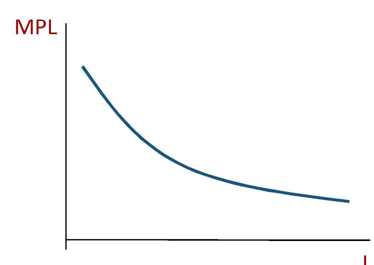
- Multiple variables
  - Partial derivative (marginal in economics)
  - $f(x,y) = x^a.y^b$
    - $f_x = \frac{\partial f(x,y)}{\partial x} = a.x^{a-1}.y^b$
    - $f_y = \frac{\partial f(x,y)}{\partial y} = b.x^a.y^{b-1}$
    - Store gradient of $f(x,y)$ into a vector

$\nabla f = \begin{bmatrix} f_x \\ f_y \end{bmatrix} = \begin{bmatrix} f_L \\ f_K \end{bmatrix} = \begin{bmatrix} A.b.K^a L^{b-1} \\ A.a.K^{a-1} L^b \end{bmatrix}$



$\mathrm{TP} = \mathrm{Y} = A.K^a L^b$

b<1



MPL

$= A.b.K^a L^{b-1}$

Examples

- $f(x) = 10x^3 + \cancel{4}$

$$\mathbb{R}_1 \to \mathbb{R}$$
$$f'(x) = 30x^2$$

- $f(x,y) = 5.x^4.y$

$$\mathbb{R}_2 \to \mathbb{R}$$
$$f_x = 20.x^3.y$$
$$f_y = 5.x^4.\underbrace{y^0}_{=1} = 5.x^4$$

Examples

- $f(x,y) = \overset{g(x)}{(1 + 5.x^4)}.\overset{f(x)}{(y - 3)}$

$$f_x = (y-3).20x^3 + (1+5.x^4).0 = (y-3).20x^3$$
$$f_y = (y-3).0 + (1+5.x^4).1 = (1+5.x^4)$$

Rule: $h(x) = g(x).f(x)$
$\rightarrow h'(x) = g'(x).f(x) + f'(x).g(x)$

Implicit Differentiation

- a technique for computing dy/dx for a function defined by an equation, accomplished by differentiating both sides of the equation (remembering to treat the variable y as a function) and solving for dy/dx
- Example: assume $f(x,y) = x^2 + y^2 = 9$, find $\frac{dy}{dx}$ without rearranging for y first:

$$\frac{d(x^2+y^2)}{dx} = \frac{d(9)}{dx}$$

$$\frac{d(x^2)}{dx} + \frac{d(y^2)}{dx} = \frac{d(9)}{dx}$$

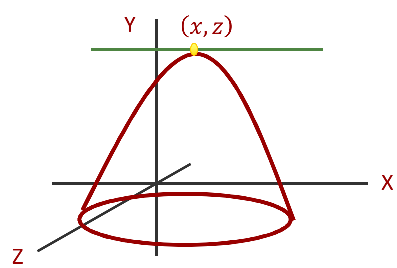
$$\frac{d(x^2)}{dx} + \frac{d(y^2)}{dy}\frac{dy}{dx} = \frac{d(9)}{dx}$$

$$2x + 2y.\frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-2x}{2y} = \frac{-x}{y}$$

Maximum and Minimum

- At a maximum or a minimum (x,z)
  - $\nabla f = \begin{bmatrix} f_x \\ f_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- Second order derivatives (Hessian)
  - $\nabla^2 F = \begin{bmatrix} f_{xx} & f_{xz} \\ f_{zx} & f_{zz} \end{bmatrix}$
  - Hessian matrix determines concavity/convexity



Examples: second order

- $f(x,y) = 5.x^4.y$

$$f_x = 20.x^3.y$$
$$f_y = 5.x^4.y^{1-1} = 5.x^4$$

$$f_{xx} = 20.3.x^{3-1}.y = 60.x^2.y$$
$$f_{xy} = 20.x^3.y^{1-1} = 20.x^3$$
$$f_{yx} = 5.4.x^{4-1} = 20.x^3$$
$$f_{yy} = 0$$

$$\begin{tikzcd}[column sep=small] & & & f(x,y) \arrow[dl] \arrow[dr] & & & \\ & & f_x \arrow[dl] \arrow[dr] & & f_y \arrow[dl] \arrow[dr] & & \\ & f_{xx} & & f_{xy} = f_{yx} & & f_{yy} & \end{tikzcd}$$

Young's theorem

Helpful Concepts

- **First Order Conditions** (F.O.C.) and **Second Order Conditions** (S.O.C.)
  - Another way of expressing the *first derivative test* and *second derivative test* you will have seen in calculus (economist love renaming things!)
  - The first order condition determines if the objective function has a local extremum (max or min)
  - The second order condition determines if this extremum is a local minimum or maximum
  - *Note*: The objective function must be continuous and twice differentiable for both conditions to hold

Example

- Find max or min of:
  - $5.x^2 + y^2 = 0$

$$f_x = 10.x = 0$$
$$f_y = 2.y = 0$$

  - Solution: (x,y) = (0,0)
- Max or min? second derivatives

$$f_{xx} = 10$$
$$f_{xy} = 0$$
$$f_{yx} = 0$$
$$f_{yy} = 2$$

Second order derivatives (Hessian)

- Positive definite – concave up (minimum)
  - If all the determinants of the minors are positive then positive definite
  - E.g., Hessian of form: $\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}$
  - $a_{11}, \begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix}$ and $\begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix} > 0$
- Negative definite – concave down (maximum)
  - If determinants of the minors switch signs
  - E.g., Hessian of form: $\begin{bmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{bmatrix}$
  - $f_{11} < 0$
  - $\begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix} > 0$
  - $\begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix} < 0$

- Minors are related to eigenvalues
- If neither positive or negative could be a saddle point: eigenvalues go in different directions.

Example

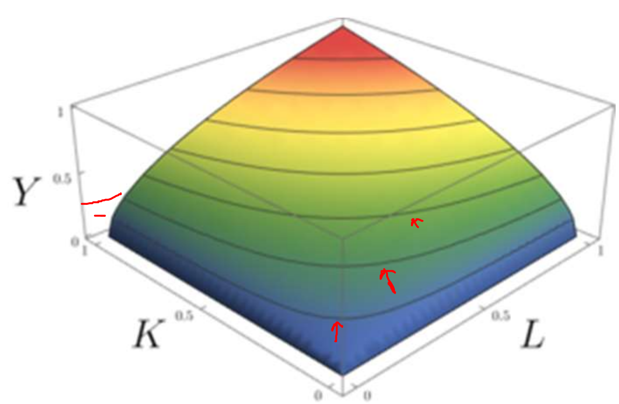
- Max or min?

$$f_{xx} = 10$$
$$f_{xy} = 0$$
$$f_{yx} = 0$$
$$f_{yy} = 2$$

  – Hessian: $\begin{bmatrix} 10 & 0 \\ 0 & 2 \end{bmatrix}$
  – Positive definite - minimum

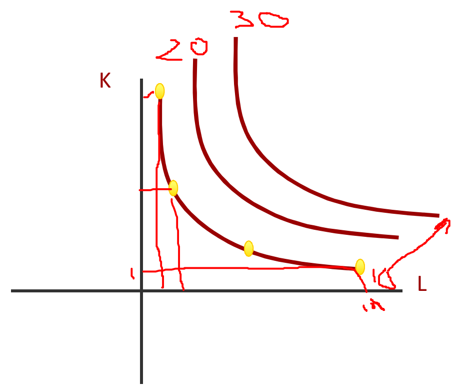
Cobb Douglas

- Used for utility and production functions
  – $f(x_1, x_2) = {x_1}^{\alpha}.{x_2}^{\beta}$
  – $f(K, L) = A.K^{\alpha}.L^{\beta}$
- Why?
  – Diminishing returns



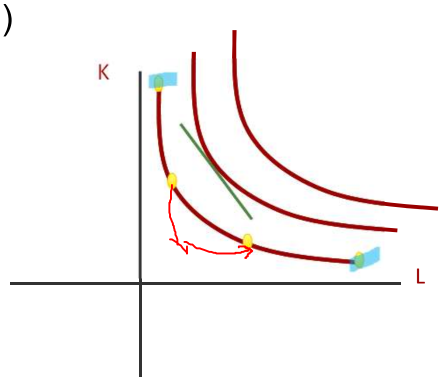
Level sets

- Combinations of K and L that give you a given amount (e.g., $f(K,L) = 10$)
  - $A = 1, \alpha = \beta = \frac{1}{2}$
  - $f(K,L) = 1.K^{1/2}.L^{1/2} = 10$
  - $K = 1, L = 10^2 = 100$
  - $L = 1, K = 10^2 = 100$
  - $K = 50, L = \left(\frac{10}{\sqrt{50}}\right)^2 = 2$



Level sets

- Notice that along a level set ($Y = f(K,L)$)
  - $\Delta\big(f(K,L)\big) = \Delta Y = f_K.\Delta K + f_L.\Delta L = 0$
- Production functions: isoquants
  - Slope is the MRTS
  - $MRTS = \frac{\Delta K}{\Delta L} = -\frac{f_L}{f_K} = -\frac{\frac{dY}{dL}}{\frac{dY}{dK}} = -\frac{MPL}{MPK}$
- Utility functions: indifference curves
  - Slope is MRS
  - $MRS = \frac{\Delta x1}{\Delta x2} = -\frac{MU_{x2}}{MU_{x1}} = -f_2/f_1$



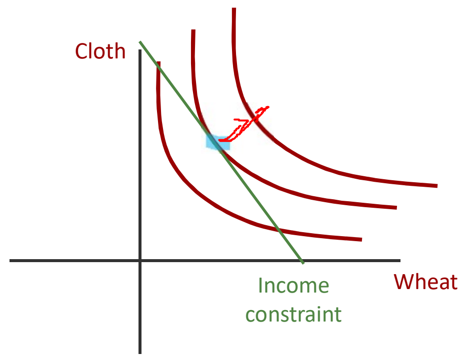
Constrained optimization

- Maximize utility st income constraint

$$U = {X_C}^a . {X_W}^b$$

Subject to (s.t.): $Y = P_c . X_C + P_W . X_W$

$$\mathcal{L} = {X_C}^a . {X_W}^b - \lambda(P_c . X_C + P_W . X_W - Y)$$



Constrained Optimization – Lagrangian Method

- Workhorse of economic optimization problems
- Technique is used to find the local minima/maxima of multivariate functions
- **Key**: add/subtract **λ** (*Lagrange multiplier*) times the constraint. Then differentiate with respect to X_C, X_W and **λ**. Finally, solve the resulting system of 3 equations

$U = {X_C}^a . {X_W}^b$ ← *Objective function*

Subject to (s.t.): $Y = P_c . X_C + P_W . X_W$ ← *Constraint function*

$\mathcal{L} = {X_C}^a . {X_W}^b - \lambda(P_c . X_C + P_W . X_W - Y)$ ← *Resulting lagrangian function*

Cobb Douglas Demand functions

$$\mathcal{L} = X_C{}^a . X_W{}^b - \lambda(P_c . X_C + P_W . X_W - Y)$$

- First order conditions:

$$- \frac{\delta \mathcal{L}}{\delta X_C} = a . X_C{}^{a-\mathbf{1}} . X_W{}^b - \boldsymbol{\lambda}(\boldsymbol{P_c}) = \mathbf{0}$$

$$- \frac{\delta \mathcal{L}}{\delta X_W} = b . X_C{}^a . X_W{}^{b-\mathbf{1}} - \boldsymbol{\lambda}(\boldsymbol{P_W}) = \mathbf{0}$$

$$- \frac{\delta \mathcal{L}}{\delta \lambda} = \boldsymbol{P_c} . X_C + \boldsymbol{P_W} . X_W - \boldsymbol{Y} = \mathbf{0}$$

Cobb Douglas Demand functions

$$\mathcal{L} = X_C{}^a . X_W{}^b - \lambda(P_c . X_C + P_W . X_W - Y)$$

- First order conditions:

$$- \frac{\delta \mathcal{L}}{\delta X_C} \Rightarrow a . X_C{}^{a-\mathbf{1}} . X_W{}^b = \cancel{\boldsymbol{\lambda}}(\boldsymbol{P_c})$$

$$- \frac{\delta \mathcal{L}}{\delta X_W} \Rightarrow b . X_C{}^a . X_W{}^{b-\mathbf{1}} = \cancel{\boldsymbol{\lambda}}(\boldsymbol{P_W})$$

$$- \frac{\delta \mathcal{L}}{\delta \lambda} = \boldsymbol{P_c} . X_C + \boldsymbol{P_W} . X_W - \boldsymbol{Y} = \mathbf{0}$$

Cobb Douglas Demand functions

$$\frac{MU_c}{MU_w} = \frac{a.X_C{}^{a-1}.X_W{}^b}{b.X_C{}^a.X_W{}^{b-1}} = \frac{\boldsymbol{P_c}}{\boldsymbol{P_w}}$$

- Simplify LHS: $\frac{a.X_W{}^b.X_W{}^{-b+1}}{b.X_C{}^a.X_C{}^{-a+1}} = \frac{a.X_W{}^{b-b+1}}{b.X_C{}^{a-a+1}}$

$$\frac{a.X_W}{b.X_C} = \frac{\boldsymbol{P_c}}{\boldsymbol{P_W}}$$

- Next rearrange for W and then C to get them as functions of the other:
  - $X_W = \frac{b.\boldsymbol{P_c}}{a.\boldsymbol{P_W}}.X_C$
  - $X_C = \frac{a.P_W}{b.P_C}.X_W$

Cobb Douglas Demand functions

- Then we plug each into the budget constraint
- Plug $X_W = \frac{b.\boldsymbol{P_c}}{a.\boldsymbol{P_W}}.X_C$ into budget constraint to solve for the demand for X_C :
  - $\boldsymbol{Y} = \boldsymbol{P_c}.X_C + \boldsymbol{P_W}.X_W$
  - $\boldsymbol{Y} = \boldsymbol{P_c}.X_C + \boldsymbol{P_W}.\frac{b.\boldsymbol{P_c}}{a.\boldsymbol{P_W}}.X_C$
  - $\boldsymbol{Y} = \boldsymbol{P_c}.X_C\left[1 + \boldsymbol{P_W}.\frac{b}{a.\boldsymbol{P_W}}\right]$
  - $\boldsymbol{Y} = \boldsymbol{P_c}.X_C.\left[1 + \frac{b}{a}\right] = \boldsymbol{P_c}.X_C.\left[\frac{a}{a} + \frac{b}{a}\right]$
  - $X_C = \frac{Y}{\boldsymbol{P_c}}\left[\frac{a}{a+b}\right]$ or $\frac{X_C.\boldsymbol{P_c}}{Y} = \left[\frac{a}{a+b}\right]$
  - This is why a and b are often referred to as the shares.

Cobb Douglas Demand functions

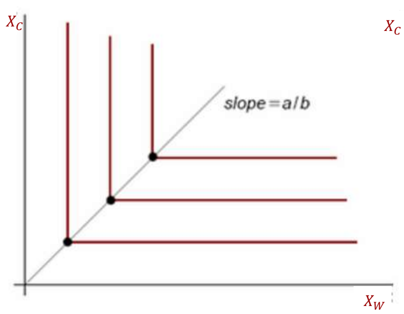
- Plug $X_C = \frac{a.\boldsymbol{P}_W}{b.\boldsymbol{P}_C}.X_W$ into the budget constraint to obtain the demand curve for X_W

$$\frac{X_W.P_W}{Y} = \left[\frac{b}{a+b}\right]$$

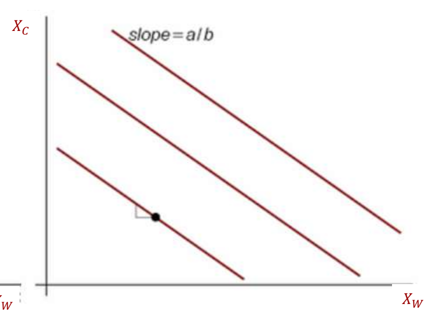
or $$X_W = \frac{Y}{\boldsymbol{P}_W}\left[\frac{b}{a+b}\right]$$

- May also define a composite price to decompose Y into real income and P:
  - $X_C = \frac{X.P}{P_C}\left[\frac{a}{a+b}\right]$ and $X_W = \frac{X.P}{P_W}\left[\frac{b}{a+b}\right]$
  - Emphasizes that it is relative prices that matter. Since $P = S_C.P_C + S_W.P_W$

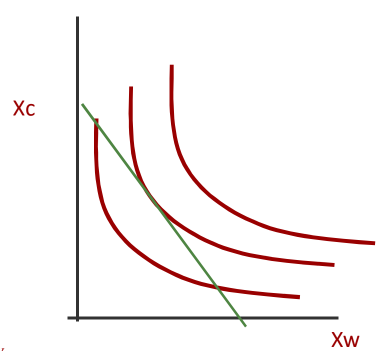
Other demand functions



Leontief fixed proportions
$U(X_W, X_C) = \min(a.X_W, b.X_C)$



Perfect Substitutes
$U(X_W, X_C) = a.X_W + b.X_C$



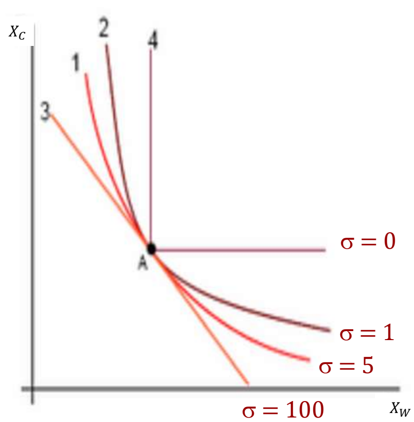
Constant elasticity of substitution
$U(X_W, X_C) = (X_W{}^{-\rho} + X_C{}^{-\rho})^{-1/\rho}$

Elasticity

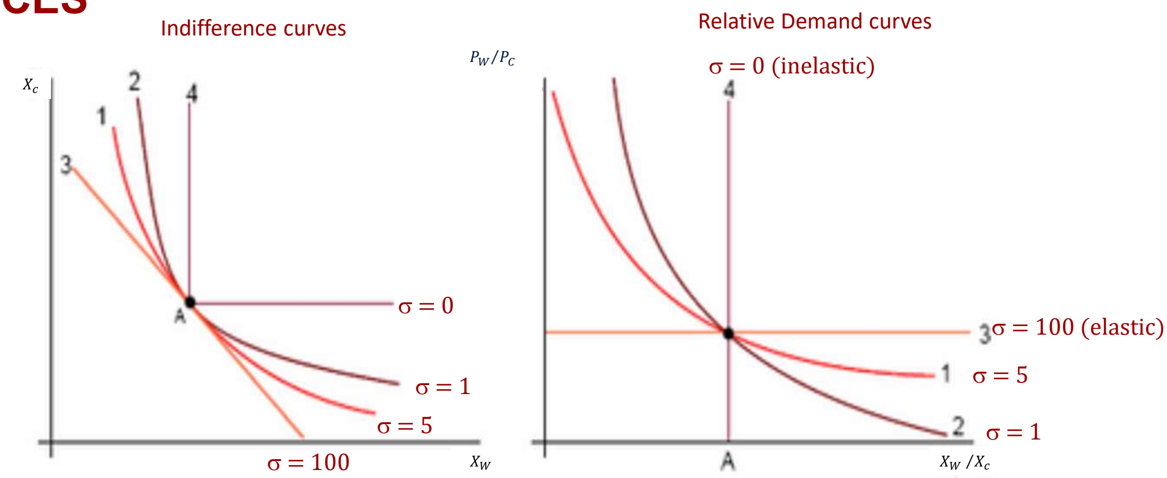
$$\frac{\Delta Y}{\Delta X}.\frac{X}{Y} = \frac{\%\ change\ in\ Y}{\%\ change\ in\ X}$$

- Sensitivity of Y to changes in X
  - e.g., sensitivity of demand Q to price changes P. When the price of a good increases, we expect demand to fall.
- In a linear function: $Q = \beta_0 - \beta_1.P$
  - elasticity = $\frac{\Delta Q}{\Delta P}.\frac{P}{Q} = \beta_1.\frac{P}{\beta_0+\beta_1.P}$
  - notice that elasticity depends on P and is therefore not constant along a straight line.
- For log functions: $log(Q) = \beta_0 - \beta_1.P$
  - elasticity = $\frac{\Delta Q}{\Delta P}.\frac{P}{Q} = \frac{\Delta logQ}{\Delta logP} = \beta_1$
  - β_1 is constant
- For Cobb Douglas: Elasticity = 1 (prove for yourself)

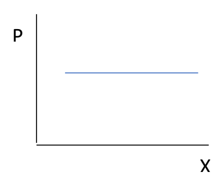
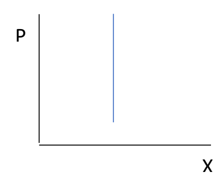
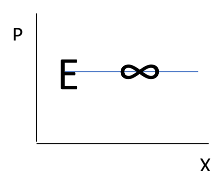
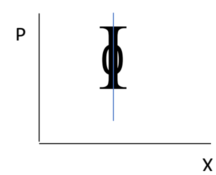
CES Indifference curves



CES



How to remember elasticity and slopes

	Elastic Elasticity > 1 X is very sensitive to changes in P		Inelastic (elasticity <1) X is not sensitive to changes in P
			

Homogeneity

- A function is homogeneous to degree k if:
  - $f(s.x_1, s.x_2, s.x_3, \dots . s.x_n) = s^k . f(x_1, x_2, x_3, \dots . x_n)$
  - e.g., $\mathrm{Q} = \mathrm{A}.K^2.L^3$

$$\mathrm{A}.(sK)^2.(sL)^3$$
$$s^2 s^3.\mathrm{A}.(K)^2.(L)^3$$
$$s^5.\mathrm{A}.(K)^2.(L)^3$$

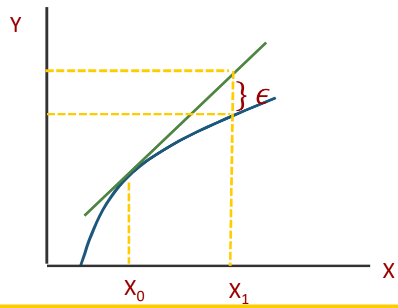
    - Is homogenous to degree 5 (increasing returns to scale)
- Other examples:
  - e.g., $\mathrm{Q} = \mathrm{A}.K^{2/3}.L^{1/3}$ is homogeneous to degree 1 (constant returns to scale)
  - e.g., $\mathrm{Q} = \mathrm{A}.K^{1/4}.L^{1/4}$ is homogeneous to degree 1/2 (decreasing returns to scale)

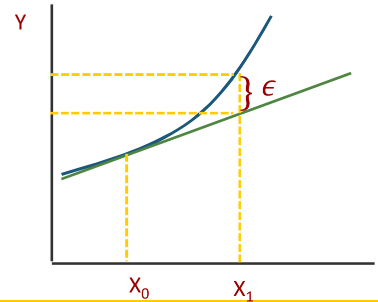
Euler's theorem

- If $f(x)$, where $x \in \mathbb{R}^n_+$ is homogeneous of degree k
  - Then $f_1.x_1 + f_2.x_2 + \dots + f_n.x_n = k.f(x)$
  - Hence if $k = 1$
  - $\sum_{i=1}^{n} f_i x_i = f(x)$
- Hence for a production function $\mathrm{Q} = \mathrm{A}.K^{2/3}.L^{1/3}$ that is homogeneous to degree 1:
  - $\mathrm{Q} = \frac{dQ}{dK}.K + \frac{dQ}{dL}.\mathrm{L} = MPK.K + MPL.\mathrm{L}$
  - Which means that all product Q are distributed to the inputs in production (no surplus or deficit)

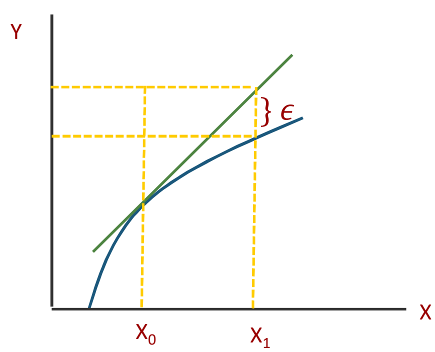
Taylor Series expansion/approximation

- If we were to use the derivative to find the Δy due to a Δx from x_0 to x_1. There is likely to be an error ϵ.
  - If function is concave then this method leads to an overestimate
  - If Convex, an underestimate
  - Conversely measuring the error tells us what type of function we have.



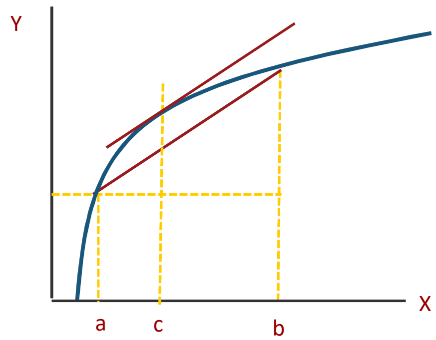


Taylor Series expansion



- Taylor series expansion corrects for the error:
  - $f(x_1) = f(x_0) + \sum_{k=1}^{n-1}\left[\frac{f^k(x_o)\,(x_1-x_o)^k}{k!}\right] + R_n$
    - Where: R_n is the remainder $R_n = f^n(\xi)(x_1 - x_0)^n/n!$ and ξ is a value that lies between x_1 and x_0
    - As $k \to \infty$, the remainder $\to 0$.
    - Note k=10 is usually large enough to get a solution within 6 decimal places.
  - Assuming k=2, the function can be rearranged to show that R_n is the error
    - $\Delta y = dy + f^2(\xi)(x_1 - x_0)^2/2$

Taylor Series expansion



- Assuming k=1, we can also obtain the mean value theorem:
  - $f(x_1) = f(x_0) + f'(\xi)(x_1 - x_0)^1$
  - $f'(\xi) = \frac{f(x_1)-f(x_0)}{x_1-x_0} = f'(x_M)$
- **Mean value theorem** is that if f is a continuous function on a closed interval (a,b) then there exists a point c (between a and b) such that the tangent at c is parallel to the secant line through the endpoints, i.e.,
  - $f'(c) = \frac{f(b)-f(a)}{b-a} = f'(c)$

Envelope Theorem

- Used when our model depends on some endogenous (X_1, X_2) and exogenous (a) variables

$$\mathcal{L} = f(X_1, X_2; a) + \lambda\big(g(X_1, X_2; a)\big)$$

- Can write a value function: $\mathrm{V}(a) = f(X_1(\mathrm{a}), X_2(a); a)$

$$\mathcal{L} = f(X_1(a), X_2(a); a) + \lambda\big(g(X_1(a), X_2(a); a)\big)$$

$$\frac{\mathrm{d}\mathcal{L}}{\mathrm{da}} = (f_1 + \lambda g_1).\frac{\mathrm{d}X_1}{da} + (f_2 + \lambda g_2).\frac{\mathrm{d}X_2}{da} - g.\frac{d\lambda}{\mathrm{da}} + (f_a + \lambda g_a)$$

- At the optimal point $f_i + \lambda g_i = 0$ and $g = 0$, hence:

$$\frac{\mathrm{d}\mathcal{L}}{\mathrm{da}} = (f_a + \lambda g_a)$$

Appendix: Helpful Definitions

- **Chain Rule** - the chain rule defines the derivative of a composite function as the derivative of the outer function evaluated at the inner function times the derivative of the inner function
- **Derivative** – the slope of the tangent line to a function at a point, calculated by taking the limit of the difference quotient, is the derivative
- **Differentiation** – the process of taking a derivative
- **Differentiable function** - a function for which $f'(x)$ exists
- **Higher-order derivative** – a derivative of a derivative, from the second derivative to the *n*th derivative, is called a higher-order derivative
- **Implicit Differentiation** – is a technique for computing dy/dx for a function defined by an equation, accomplished by differentiating both sides of the equation (remembering to treat the variable y as a function) and solving for dy/dx
- **Lagrange Multiplier/Method** – workhorse technique to solve optimization problems with multiple variables
- **Logarithmic Differentiation** - technique that allows us to differentiate a function by first taking the natural logarithm of both sides of an equation, applying properties of logarithms to simplify the equation, and differentiating implicitly